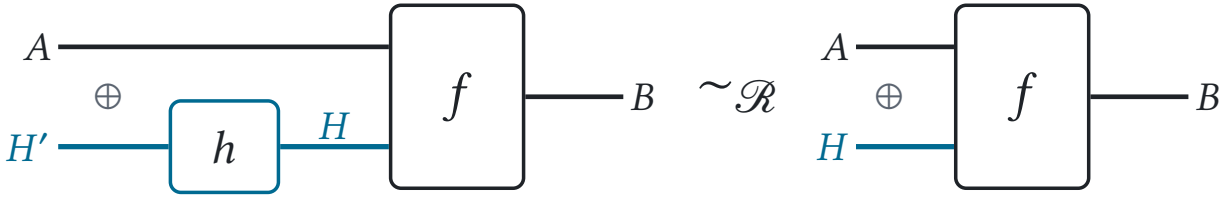
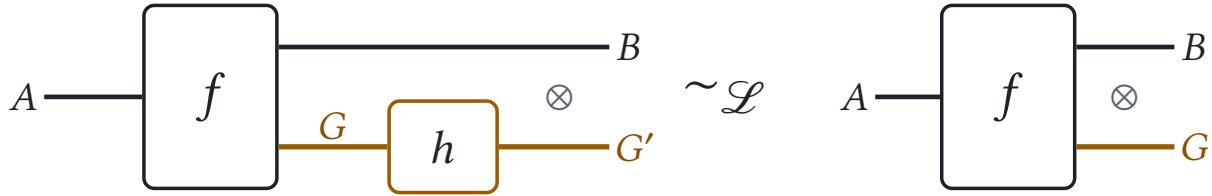
(a) $\mathscr{R}[\mathscr{C}]$: a heap enters beside the input, and only access to it counts



normal form: $A \to A \oplus H$, then f;
$\mathscr{R}[\text{Unitary}] \simeq \text{Isometry}$

(b) $\mathscr{L}[\mathscr{C}]$: garbage leaves beside the output, and its contents do not count



normal form: f, then $B \otimes G \to B$;
$\mathscr{L}[\mathscr{R}[\text{Unitary}]] \simeq \text{CPTP}$